Pranav H. Premnath

Department of Physics & Astronomy
University of California, Irvine

premnatp@uci.edu

ADS · ORCID · Personal Website

Research Interests

Exoplanets • Stellar Astrophysics • Radial Velocities • Transit Photometry • Exoplanet Occurrence Rates • Bayesian Statistics

Education

University of California, Irvine 2022–Present
Ph.D. Physics (Concentration in Astronomy and Astrophysics), expected 2027
Advisor: Paul Robertson
M.S. Physics, 2024

The University of Texas at Austin 2017–2021
B.S. Astronomy
B.S. Physics
Major GPA: 3.71 Overall GPA: 3.69

Research Experience

University of California, Irvine 2022–Present
Graduate Student Researcher
Advisor: Paul Robertson

SETI Institute / Hat Creek Radio Observatory 2021–2022
Research Assistant
Advisor: Wael Farah

University of Texas at Austin 2020–2021
Undergraduate Researcher
Advisor: Justin Spilker and Caitlin Casey

University of Texas at Austin 2019–2020
Undergraduate Researcher
Advisor: Ya-Lin Wu and Brendan Bowler

Telescope Time Awarded

- **PI** — "Transit Surface Reconstruction of HD 189733 with KPF"
  Keck Planet Finder (KPF), W. M. Keck Observatory
  2024A 2 half-nights awarded
- **PI** — "Transit Surface Reconstruction of HD 209458 and HD 189733 with KPF"
  Keck Planet Finder (KPF), W. M. Keck Observatory

2024B — 5 half-nights awarded

- **PI** — “Characterizing Starspot Spectra via Spot Crossings with KPF”
  Keck Planet Finder (KPF), W. M. Keck Observatory
  2026A — 3 half-nights awarded

- **PI** — “Characterizing Starspots through Spot Crossings with MuSCAT”
  MuSCAT, Las Cumbres Observatory, Haleakala
  2026A — 2 half-nights awarded

- **PI** — “Characterizing Starspots via Spot Crossings with WIRC”
  WIRC, Palomar Observatory
  2026A — 1 night awarded

- **Co-I** — “Mapping Stellar Activity with Star Spot Crossing Events.”
  ARCTIC, Apache Point Observatory
  2025-2026 — 12 half-nights awarded

Observing Experience

- Allen Telescope Array — 100+ nights
- Keck I – KPF — 46 nights
- Apache Point Observatory – ARCTIC — 11 nights
- Keck I – HIRES — 6 nights
- Las Campañas Observatory – WINERED — 5 nights
- Lick Observatory – Shane AO — 3 nights
- Keck II – NIRSPEC — 2 nights
- Palomar Observatory – WIRC — 1 night

Publications

First-author Publications

1. **Premnath** et al. 2026, “Searching for GEMS: Three warm Saturns and a super-Jupiter orbiting four early M-dwarfs", *The Astronomical Journal, Volume 169, Issue 2, id.149, 92 pp.*
2. **Premnath** et al. 2020, “Evidence for a Buried AGN in an Extremely Bright Dusty Galaxy at z = 2", *Research Notes of the AAS, Volume 4, Issue 10, id.173.*
3. **Premnath** et al. 2020, “Dynamical Masses of Young Stars Inferred from Two Transitions of CO with ALMA", *Research Notes of the AAS, Volume 4, Issue 7, id.100.*

Co-authored Publications

1. Wright et al. 2026, “A Simple One-free-parameter Model of the Solar Chromosphere Explains Solar Chromospheric Line Variations Measured by NEID at WIYN during the 2023 October Solar Eclipse", *Research Notes of the AAS, Volume 10, Issue 1, id.14.*

2. Handley et al. 2026, “The KPF-SLOPE Survey - Small, Compact Multi-Planet Systems Appear Spin-Orbit Aligned", *arXiv:2603.23713*

3. Beard et al. 2025, “Jitter Across 15 yr: Leveraging Precise Photometry from Kepler and TESS to Extract Exoplanets from Radial Velocity Time Series", *The Astronomical Journal, Volume 169, Issue 2, id.149, 92 pp.*

4. Crossfield et al. 2025, “OrCAS: Origins, Compositions, and Atmospheres of Sub-Neptunes. I. Survey Definition", *The Astronomical Journal, Volume 169, Issue 2, id.89, 15 pp.*

5. Kroft et al. 2025, “A Pair of Dynamically Interacting Sub-Neptunes Around TOI-6054", *The Astronomical Journal, Issue 3, Volume 170, id.150, 17 pp.*

6. Teng et al. 2025, “Stellar Obliquity of the Ultra-short-period Planet System HD 93963", *The Astronomical Journal, Issue 1, Volume 170, id.51, 18 pp.*

7. Beard et al. 2024, “Utilizing Photometry from Multiple Sources to Mitigate Stellar Variability in Precise Radial Velocities: A Case Study of Kepler-21", *The Astronomical Journal, Volume 168, Issue 4, id.149, 17 pp.*

8. Joshi et al. 2024, “Wideband detection of FRB 20240114A above 2 GHz with the Allen Telescope Array", *The Astronomer’s Telegram, No. 16599.*

9. Sheikh et al. 2024, “Characterization of the repeating FRB 20220912A with the Allen Telescope Array", *MNRAS, Volume 527, Issue 4, pp.10425–10439.*

10. Bright et al. 2023, “Precise measurements of self-absorbed rising reverse shock emission from gamma-ray burst 221009A", *Nature Astronomy, Volume 7, pp.986–995.*

11. Sheikh et al. 2022, “Bright radio bursts from the active FRB 20220912A detected with the Allen Telescope Array", *The Astronomer’s Telegram, No. 15735.*

12. Perez et al. 2022, “Breakthrough Listen Search for the WOW! Signal", *Research Notes of the AAS, Volume 6, Issue 9, id.197.*

Presentations

- **Searching for GEMS: Building the Largest Volume-Limited Sample of Giant Planets around M Dwarfs**
  Emerging Researchers in Exoplanet Science Symposium XI – University of Michigan, Ann Arbor
  (Poster, July 2026)

- **Searching for GEMS: Building the Largest Volume-Limited Sample of Giant Planets around M Dwarfs**
  Sagan Summer Workshop – California Institute of Technology
  (Poster, July 2026)

- **Searching for GEMS: Three warm Saturns and a super-Jupiter orbiting four early M-dwarfs**
  WIYN Board Meeting
  (Presentation, April 2026)

- **Searching for GEMS: Characterizing Giant Planet Candidates within 200pc**
  NExScI ExSoCal Meeting – University of California, Los Angeles
  (Poster, December 2025)

- **Probing Stellar Surfaces Using Transiting Exoplanets**
  Know Thy Star, Know Thy Planet 2 – California Institute of Technology
  (Poster, February 2025)
- **Pilot SETI Survey with the Upgraded Allen Telescope Array**
  NExScI ExSoCal Meeting – California Institute of Technology
  (Poster, December 2023)
- **Technosignature Searches with the Allen Telescope Array***PSETI Symposium*
  PSETI Symposium – The Pennsylvania State University
  (Presentation (with W. Farah), June 2022)
- **Evidence for a Buried AGN in an Extremely Bright Dusty Galaxy at z = 2**
  237th Meeting of the American Astronomical Society
  (Poster, January 2021)
- **Dynamical Mass Measurement of a Young Star with ALMA**
  235th Meeting of the American Astronomical Society
  (Poster, January 2020)

Honors & Awards

- **Sagan Workshop Travel Award** 2026
- **Sagan Workshop Travel Award** 2025
- **UT Austin College of Natural Sciences Research Distinction Winner** 2021
- **Dr. Arnold Romberg Endowed Scholarship Fund** 2020–2021
- **Karl G. Henize Endowed Scholarship** 2020
- **John W. Cox Endowment for Advanced Studies in Astronomy** 2020
- **Darrell W. Moffitt, Jr. Endowed Presidential Scholarship** 2019–2020
- **John W. Cox Endowment for Advanced Studies in Astronomy** 2019

Teaching Experience

Teaching Assistant, University of California, Irvine

- **PHY 3B** – Basic Physics II Winter 2026
- **PHY 20A** – Introduction to Astronomy Winter 2026
- **PHY 7D** – Classical Physics Spring 2025
- **PHY 138** – Astrophysics of Galaxies Winter 2025
- **PHY 3LB** – Basic Physics Laboratory Winter 2023, Summer 2023, Winter 2024, Winter 2025
- **PHY 3A** – Basic Physics I Fall 2024
- **PHY 3C** – Basic Physics III Fall 2024
- **PHY 7LC** – Classical Physics Laboratory Fall 2023
- **PHY 61C** – Introduction to Astrophysics Spring 2023, Spring 2024

Professional Service

- Administrator, Physics Graduate Caucus, UCI (2024–2025)
- Assistant Administrator, Physics Graduate Caucus, UCI (2023–2024)
- Co-lead, Aspiring Allyship Group, UCI (2024–2025)
- Mentor, Physics and Astronomy Community Excellence, UCI (2023–Present)
- Treasurer, Astronomy Students Association, UT Austin (2019–2021)

Technical Skills

Programming: Python, Bash, Mathematica

Scientific Libraries: Astropy, NumPy, SciPy, Pandas, Matplotlib, emcee

Astronomy Packages: exoplanet, RadVel, rmfit, ironman, ANTARESS, HPF-SpecMatch, spotrod, TGLC, lightkurve, batman

Astronomy Software: AstroImageJ, Aladin, CASA

Methods: Bayesian inference, MCMC, Gaussian processes, time-series analysis

References

Dr. Paul Robertson	paul.robertson@uci.edu
Dr. Shubham Kanodia	skanodia@carnegiescience.edu
Dr. Caleb Cañas	c.canas@nasa.gov
Dr. Samuel Halverson	samuel.halverson@jpl.nasa.gov